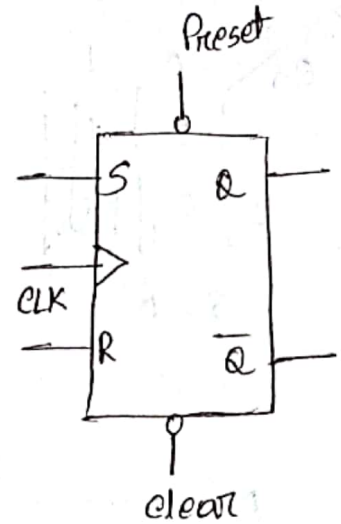


Truth table :

Preset	clear	Q_n
0	0	Not used
0	1	1
1	0	0
1	1	FF operation



SR to D Flip Flop conversion:

- Steps:
- ① Note available flip flop and required flip flop
 - ② Write characteristic table of required FF
 - ③ Write Excitation table of available FF
 - ④ Solve boolean expression
 - ⑤ Draw circuit

Here, Available Flip Flop $\rightarrow$ SR FF
 Required Flip Flop $\rightarrow$ D FF

characteristic table of required FF (DFF)

	Q_n	D	Q_{n+1}	S	R
0	0	0	0	0	X
2	0	1	1	1	0
1	1	0	0	0	1
3	1	1	1	X	0

Excitation table (SR)

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

S

D

Q_n

	0	1
0	0 ₀	1 ₂
1	0 ₁	X ₃

$$S = D$$

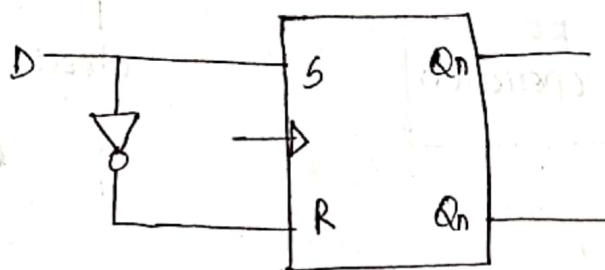
R

D

Q_n

	0	1
0	X ₀	0 ₂
1	1 ₁	0 ₃

$$R = \bar{D}$$



SR to JK flip flop conversion:

- ① Note available FF and required FF
- ② characteristic table of required FF
- ③ Write excitation table of available FF
- ④ Solve Boolean expression
- ⑤ Draw circuit

Available Flip-flop = SR FF

Required Flip-flop = JK FF

characteristic table of JK FF

$J \quad K \quad Q_n$	Q_n	J	K	Q_{n+1}	S	R
0	0	0	0	0	0	X
2	0	0	1	0	0	X
4	0	1	0	1	1	0
6	0	1	1	1	1	0
1	1	0	0	1	X	0
3	1	0	1	0	0	1
5	1	1	0	1	X	0
7	1	1	1	0	0	1

characteristic table of SR
od ~~SR~~ Excitation
Table ~~SR~~

Excitation table of SR FF

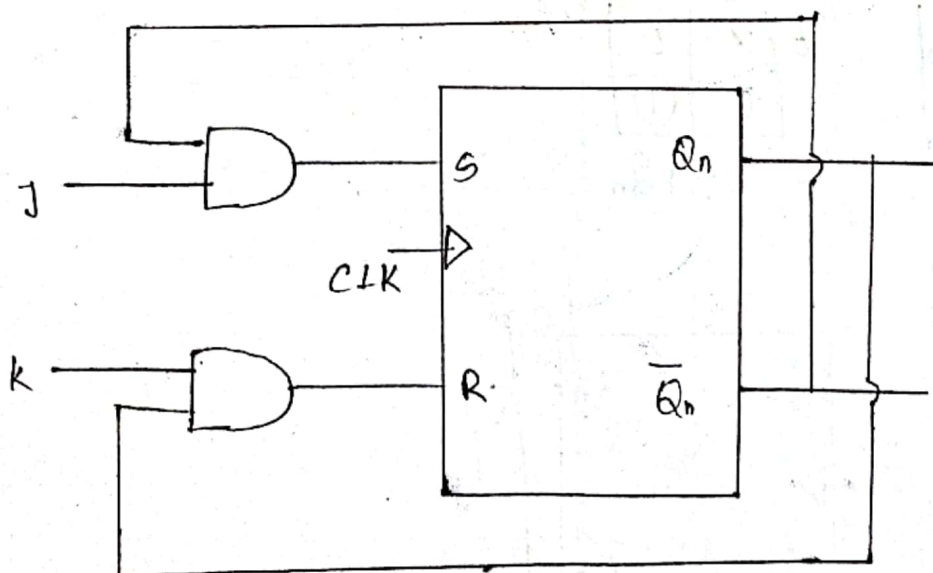
Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

S Q_n	JK			
	00	01	11	10
0	0 ₀	0 ₂	1 ₆	1 ₄
1	X ₁	0 ₃	0 ₇	X ₅

$$S = \bar{Q}_n J$$

R Q_n	JK			
	00	01	11	10
0	X ₀	X ₂	0 ₆	0 ₄
1	0 ₁	1 ₃	1 ₇	0 ₅

$$R = Q_n k$$



SR to T flip flop conversion :

- ① Note available FF and required FF
- ② Write characteristic table of required FF
- ③ Write excitation table of available FF
- ④ solve Boolean expression
- ⑤ Draw circuit.

Here, Available FF = SR FF

Required FF = T FF

Characteristic table of T FF

Q_n	T	Q_{n+1}	S	R
0	0	0	0	X
0	1	1	1	0
1	0	1	X	0
1	1	0	0	1

Excitation table of SR FF

Q_n	Q_{n+1}	S	R
0	0	0	X
0	1	1	0
1	0	0	1
1	1	X	0

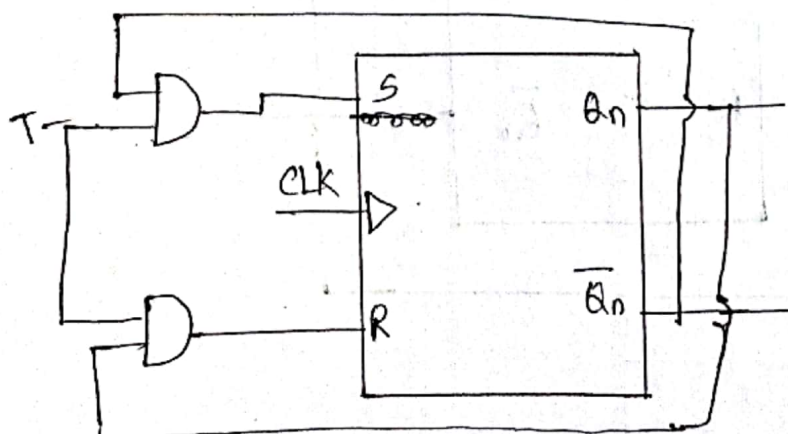
*T = 1-2/01
Qn & S opposite
2/01 & 1-2/01
1-2/01*

S	T	Q_n	0	1
0	0	0	0	① ₂
1	X	1	0	③

$$S = T \bar{Q}_n$$

R	T	Q_n	0	1
0	0	0	X	0
1	0	1	0	①

$$R = T Q_n$$



Jk to D Flip Flop Conversion:

Steps - Same as previous conversions.

Here, Available FF = JK FF

Required FF = D FF

Characteristic table of D FF

Q_n	D	Q_{n+1}	J	K
0	0	0	0	X
0	1	1	1	X
1	0	0	X	1
1	1	1	X	0

J \ D	0	1
0	0 ₀	1 ₁
1	X ₁	X ₃

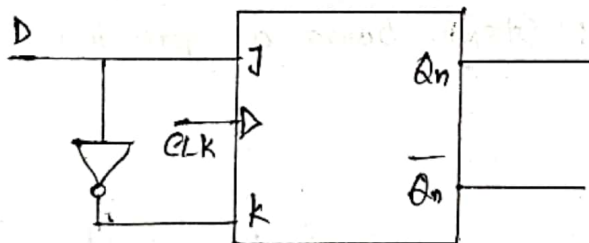
$$J = D$$

K \ D	0	1
0	X ₀	X ₂
1	1 ₁	0 ₃

$$K = \overline{D}$$

Excitation table of JK FF

Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0



Jk to T Flip Flop conversion: Steps - Same as previous conversions

Here, Available Flip Flop = JK FF

Required Flip Flop = T FF

Characteristic table T FF

Q_n	T	Q_{n+1}	J	K
0	0	0	0	X
0	1	1	1	X
1	0	1	X	0
1	1	0	X	1

Excitation table of JK FF

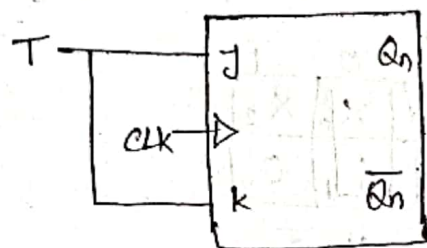
Q_n	Q_{n+1}	J	K
0	0	0	X
0	1	1	X
1	0	X	1
1	1	X	0

J \ Q_n	0	1
0	0 ₀	1 ₂
1	X ₁	X ₃

J = T

K \ Q_n	0	1
0	X ₀	X ₂
1	0 ₁	1 ₃

K = T



D to T flip-flop conversion: Steps: Same as previous conversion.

Available FF = D FF

Required FF = T FF

Characteristic table of T FF

Q_n	T	Q_{n+1}	D
0	0	0	0
0	1	1	1
1	0	1	1
1	1	0	0

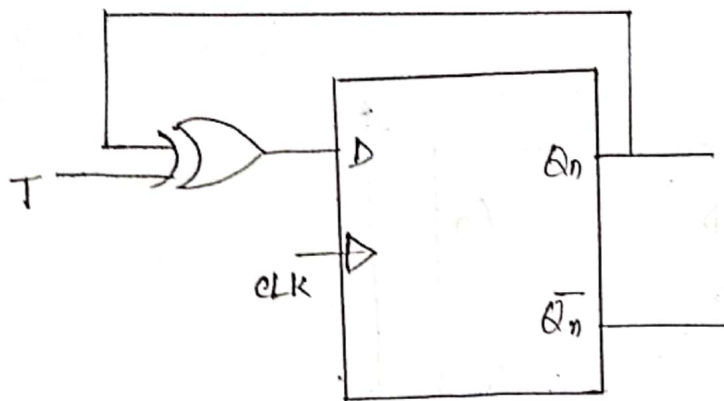
Excitation table of D FF

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

D \ Q_n	0	1
0	0 ₀	1 ₂
1	1 ₁	0 ₃

$$D = T\bar{Q}_n + \bar{T}Q_n$$

$$= T \oplus Q_n$$



D To JK flip-flop conversion: Steps - Same as previous conversion

Here, Available FF = D FF

Required FF = JK FF

Characteristic table of JK FF

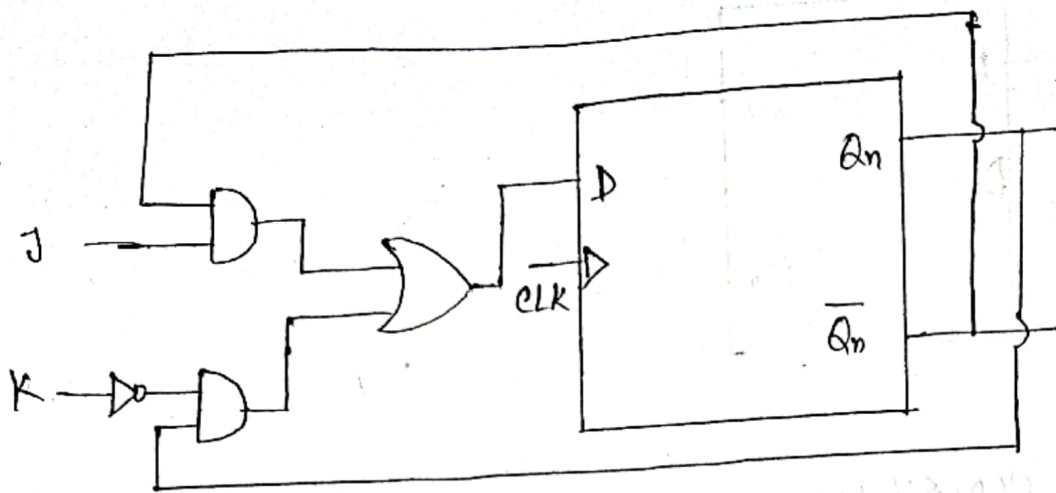
Q_n	J	K	Q_{n+1}	D
0	0	0	0	0
0	0	1	0	0
0	1	0	1	1
0	1	1	1	1
1	0	0	1	1
1	0	1	0	0
1	1	0	1	1
1	1	1	0	0

Excitation table of D FF

Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

		JK			
		00	01	11	10
Q_n	0	0 ₀	0 ₂	1 ₆	1 ₄
	1	1 ₁	0 ₃	0 ₇	1 ₅

$$D = Q_n \bar{K} + \bar{Q}_n J$$



T to D flip flop conversion: Steps - Same as previous conversion

Here, Available FF = TFF

Required FF = DFF

Characteristic table of DFF

Q_n	D	Q_{n+1}	T
0	0	0	0
0	1	1	1
1	0	0	1
1	1	1	0

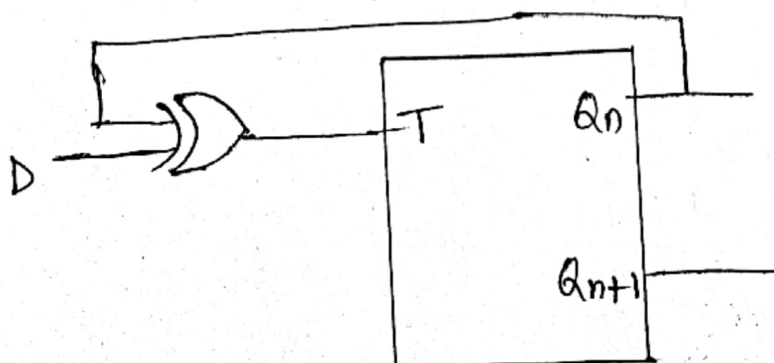
Excitation table of TFF

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

	D	T
Q_n	0	1
0	0 ₀	1 ₀
1	1 ₁	0 ₃

$$T = D\bar{Q}_n + \bar{D}Q_n$$

$$= D \oplus Q_n$$



T to JK flip-flop conversion: Steps - Same as previous conversion

Here, Available FF = T FF

Required FF = JK FF

Characteristic table of JK FF

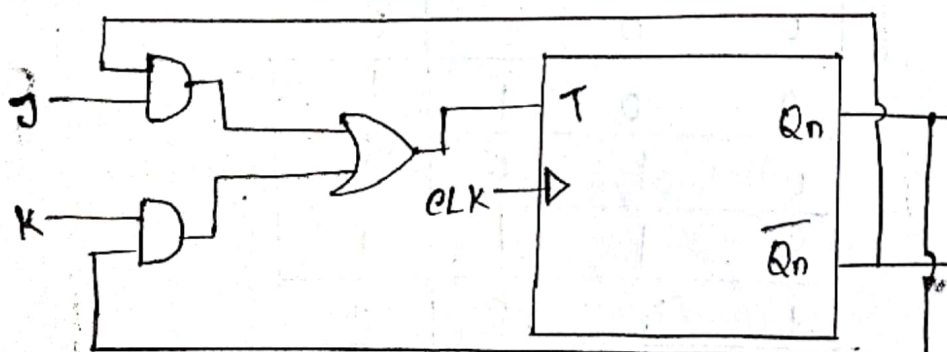
Q_n	J	K	Q_{n+1}	T
0	0	0	0	0
0	0	1	0	0
0	1	0	1	1
0	1	1	1	1
1	0	0	1	0
1	0	1	0	1
1	1	0	1	0
1	1	1	0	1

Excitation table of T FF

Q_n	Q_{n+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

		JK			
Q_n	T	00	01	11	10
		0 0	0 2	1 6	1 4
0	0	0 0	0 2	1 6	1 4
1	0	0 1	1 3	1 7	0 5

$$T = Q_n K + \bar{Q}_n J$$



JK to SR, T to SR and D to SR Flip Flop conversion.

→ when $S=1$ and $R=1$ then $Q_{n+1} = \text{Invalid}$. But for JK, T and D, $Q_{n+1} = 1$ will never go in invalid state. So we cannot form JK to SR, T to SR and D to SR Flip Flop.

Examples on Flip Flop

① A New FF is having behaviour as described below. It has two inputs X and Y and when both inputs are same and they are 1,1. The FF is going to set else FF resets. If both inputs are getting different and they are 0,1 the FF complements itself otherwise it is going to retain the last state which of the following expression is the characteristic expression of the new

FF?

(A) $XQ + Y\bar{Q}$ (B) $X\bar{Q} + YQ$ (C) $X\bar{Q} + Y\bar{Q}$ (D) None

(Truth table)

Inputs		Output
X	Y	
0	0	0
0	1	$\bar{Q}$
1	0	Q
1	1	1

(characteristic table)

Q_n	X	Y	Q_{n+1}
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	1